LESSON PLAN

BASIC RADIO & AVIONICS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Security of Data and Information in a Computer Operating in Naval Network's
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Security of Data and Information in a Computer Operating in Naval Network’s
5. Specific Objectives	22.1.1.1 Trainee will be able to explain computer operations and its applications viz hardware, software and networking.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Network Security Protocols; Safety: No live network access; Supervision: Instructor monitors PPT display
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three main components of a computer system?
(b) Define networking in the context of naval operations.
(c) Why is data security critical for naval networks?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Threats to Naval Network Data • Types of Security Attacks • Data Encryption Fundamentals • Access Control Mechanisms • Firewall and Intrusion Detection	L & Ds	PPT, IB	15 min
(b)	• Data Integrity Verification • Authentication Protocols • Secure Communication Channels • Incident Response Procedures • Role of Naval Cyber Defense	L & Ds	PPT, PDF	20 min
(c)	• Password Management Best Practices • Physical Security of Network Nodes • Regular Software Updates • Backup and Recovery Strategies • Compliance with Naval Security Directives	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between data confidentiality and data integrity.
(b) Describe one method used to prevent unauthorized access in a naval network.
(c) How does regular software updating contribute to network security?

12. Home Assignment and Follow Up	Prepare a brief report on how the concepts learned in this lesson apply to the network topology discussed in LP 10.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Network, LAN, WAN and Network Topology.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min